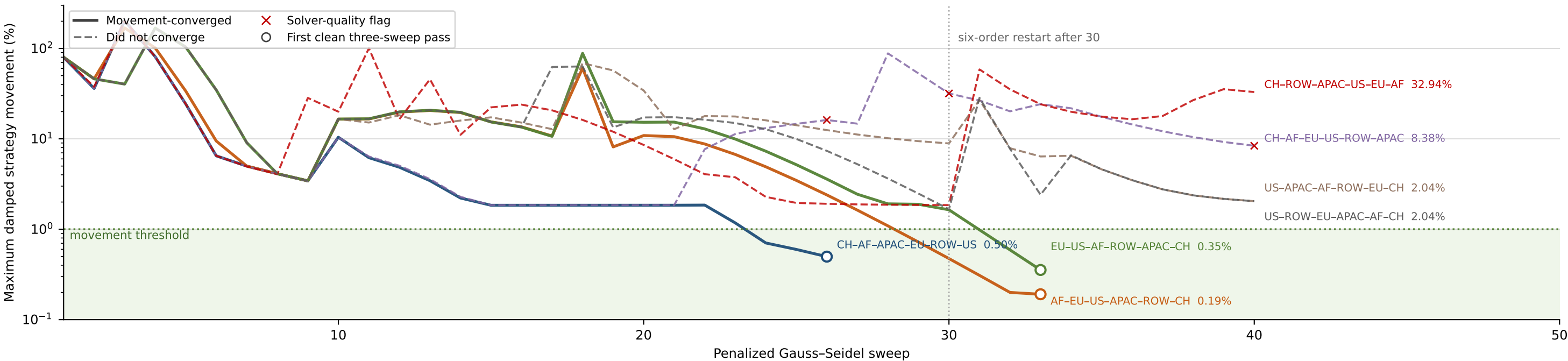


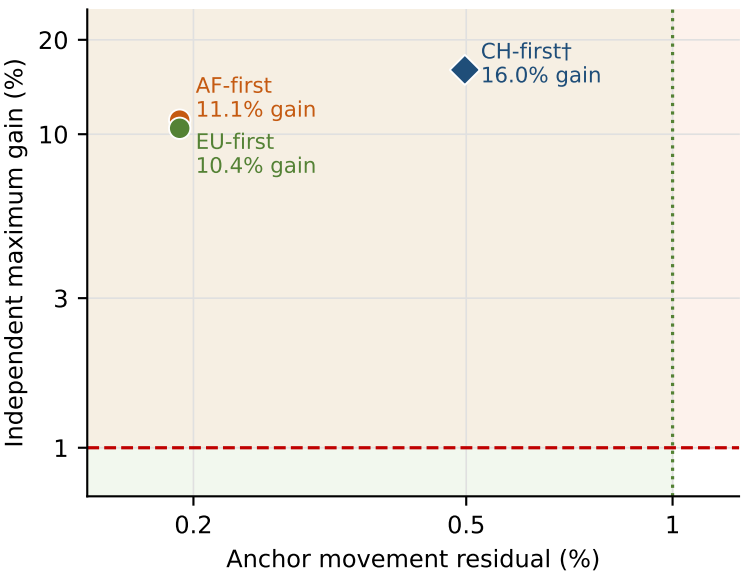
Algorithm convergence and equilibrium verification

Penalized movement locates stable basins; frozen-profile zero-proximal regret determines acceptance.

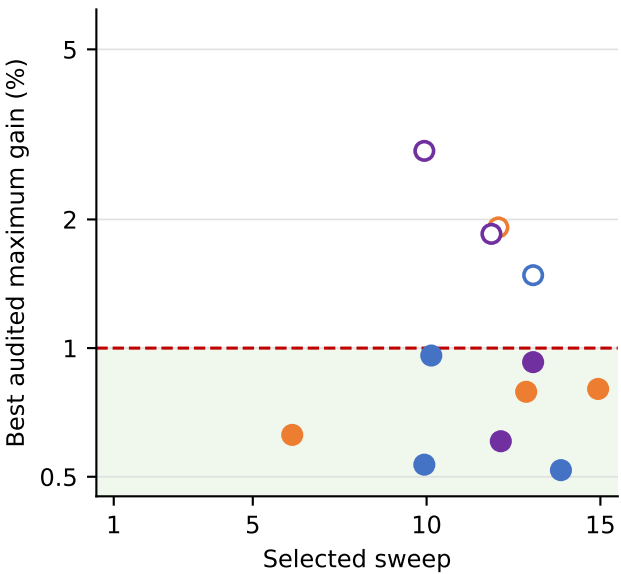
A Penalized and damped basin discovery: three of seven reach the 1% stopping rule



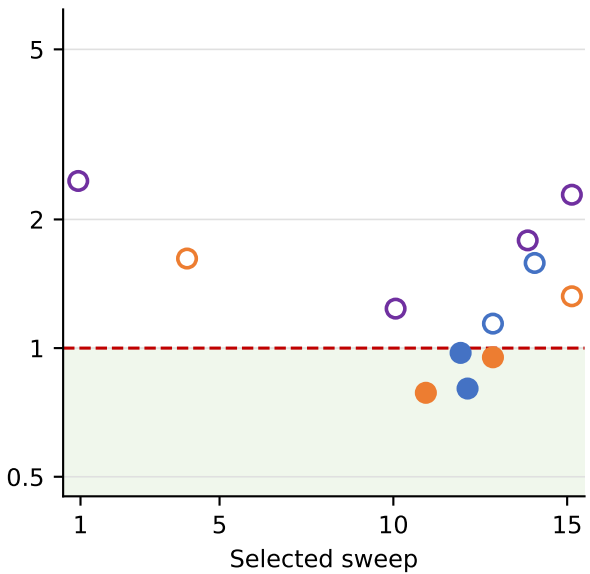
B Stable movement did not imply equilibrium



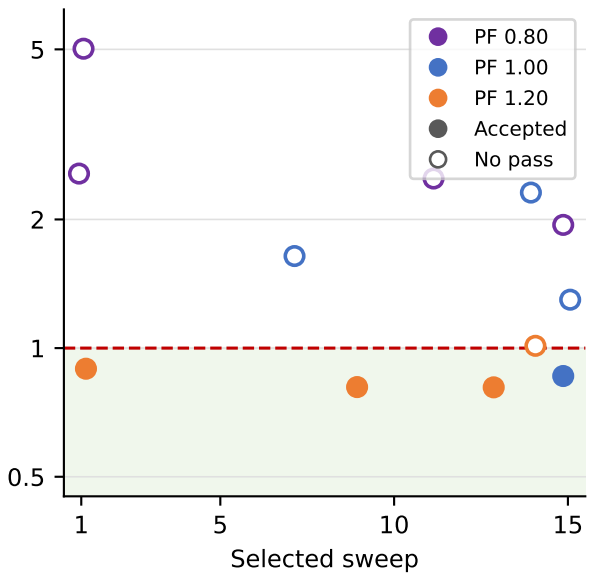
C CH-first: 8/12 pass



D AF-first: 4/12 pass



E EU-first: 4/12 pass



Movement pass: residual $\leq 1\%$ for three consecutive solver-clean sweeps. Audit pass: all six best responses succeed and maximum gain $\leq 1\%$. † CH-first direct point is the documented three-start maximum; AF/EU direct points are one-start audits. Branch panels show the first passing sweep, or the best audited sweep if none passed.